

HICKS BAY, New ZealandWMO: **931960**Lat: **37.5611S**Lon: **178.3144E**Elev: **152**StdP: **14.62**Time Zone: **12.00 (NZL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	40.8	42.7	31.5	25.9	44.5	33.9	28.7	45.9	41.3	55.9	38.6	54.9	9.3	220	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	74.1	67.6	72.7	66.6	71.5	65.9	69.6	72.0	68.4	71.0	67.4	70.1	15.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.7	106.1	70.9	67.4	101.4	70.1	66.3	97.4	69.2	33.8	71.9	32.8	71.2	31.9	70.2	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.7	32.6	29.3	DB	37.1	76.8	1.1	2.3	36.3	78.4	35.6	79.8	35.0	81.1	34.2	82.7	(4)
(5)			WB	34.1	71.1	1.4	1.5	33.1	72.2	32.2	73.0	31.4	73.8	30.4	74.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	65.3	66.5	64.5	61.2	57.6	53.7	52.5	52.7	54.5	56.6	58.9	62.8	(6)	
	DBStd	5.88	3.10	2.86	3.20	3.67	3.89	3.88	3.51	3.19	3.12	3.07	3.27	3.21	(7)	
	HDD50	43	0	0	0	0	1	10	15	12	4	1	0	0	(8)	
	HDD65	2395	34	16	48	120	231	340	387	380	314	261	183	81	(9)	
	CDD50	3280	475	461	449	337	236	120	94	97	140	206	268	397	(10)	
	CDD65	154	44	57	32	7	1	0	0	0	0	0	1	12	(11)	
	CDH74	42	13	23	5	1	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	15.1	14.7	14.4	14.3	14.1	14.7	15.3	15.4	15.1	16.1	16.4	16.2	14.5	(14)	
(15) Precipitation	PrecAvg	70.9	4.4	4.5	5.3	5.8	6.7	8.0	8.0	7.4	6.0	4.9	4.0	4.9	(15)	
	PrecMax	86.4	11.8	12.2	14.7	13.0	13.4	12.9	14.9	12.8	10.2	10.7	8.0	13.7	(16)	
	PrecMin	56.6	0.7	0.5	1.5	3.1	1.5	3.3	2.1	5.0	3.6	2.0	1.4	1.3	(17)	
	PrecStd	8.5	2.6	3.2	2.9	2.3	3.1	2.7	2.8	2.1	1.6	2.2	1.7	3.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.5	76.3	74.6	71.4	68.1	63.8	62.0	61.6	64.2	65.7	70.0	72.6	(19)	
		MCWB	68.6	69.2	67.4	65.6	62.3	60.4	58.4	56.9	57.9	59.6	63.8	66.3	(20)	
	2%	DB	73.7	74.5	72.7	69.5	66.1	62.5	60.6	60.2	62.1	64.2	67.6	70.8	(21)	
		MCWB	67.5	68.1	66.3	64.6	62.4	59.3	57.2	56.1	57.3	58.9	62.3	65.4	(22)	
	5%	DB	72.2	73.0	71.3	68.2	64.8	61.3	59.4	59.1	60.8	63.0	65.9	69.4	(23)	
		MCWB	66.5	66.9	65.5	63.5	61.3	58.1	55.8	55.5	56.6	58.2	60.9	64.5	(24)	
	10%	DB	70.8	71.6	69.9	66.9	63.8	60.1	58.4	58.2	59.7	61.9	64.4	68.1	(25)	
		MCWB	65.6	66.0	64.9	62.6	60.2	56.8	54.7	54.6	55.9	57.5	59.9	63.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	71.3	69.8	68.6	65.4	62.1	60.2	59.4	60.3	61.9	65.3	68.3	(27)	
		MCDB	73.5	73.6	72.0	70.1	66.2	62.8	61.3	60.2	61.7	63.5	68.2	70.3	(28)	
	2%	WB	69.2	70.0	68.2	66.4	63.8	60.7	58.4	58.1	59.2	60.6	63.5	66.9	(29)	
		MCDB	72.0	72.2	70.6	68.1	65.0	61.9	59.8	59.2	60.7	62.7	66.0	69.0	(30)	
	5%	WB	67.9	68.9	67.1	65.1	62.5	59.3	57.0	56.9	58.1	59.6	62.1	65.8	(31)	
		MCDB	70.8	71.4	69.9	67.0	64.2	60.8	58.7	58.5	59.9	61.7	64.6	68.0	(32)	
	10%	WB	66.7	67.7	65.9	63.7	60.9	57.6	55.5	55.5	57.0	58.5	61.0	64.7	(33)	
		MCDB	69.7	70.6	69.2	66.2	63.5	59.7	57.8	57.5	59.0	61.0	63.7	67.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.4	9.7	9.8	9.5	9.6	9.2	9.6	9.4	9.5	9.4	9.0	(35)	
		MCDBR	10.5	10.5	10.8	9.8	9.4	8.5	8.8	9.6	9.6	10.1	10.4	9.9	(36)	
	5% WB	MCWBR	7.3	7.0	7.3	7.2	7.3	7.6	7.4	7.8	7.6	7.9	8.0	7.3	(37)	
		MCWBR	8.7	8.3	8.7	7.4	6.5	6.8	7.0	7.3	7.5	8.5	9.2	8.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.350	0.346	0.332	0.318	0.307	0.307	0.303	0.305	0.328	0.340	0.339	0.346	(40)	
		taud	2.568	2.599	2.634	2.661	2.661	2.631	2.644	2.643	2.556	2.531	2.551	2.577	(41)	
	Ebn at Noon		312	306	298	284	268	258	266	284	294	304	313	314	(42)	
		Edh at Noon	33	31	28	24	21	20	21	24	30	33	34	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2174	1833	1487	1079	770	610	666	946	1309	1756	2000	2111	(44)	
	RadStd		128	115	110	63	44	42	38	56	109	105	118	143	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.63	N/A	N/A	+1.08	+1.28	+1.19	N/A	N/A	N/A	+52
(47) Regional (2 neighbors)		+0.58	+0.59	+0.77	+0.81	N/A	N/A	-20	-144	+191	+52

Nomenclature: See separate page